

GE Healthcare

Functool 9

Installation/Service Manual



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Revision History

Date	Rev	Change Description
July-2010	1	Initial Release

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation *Damage in shipment* written on **all** copies of the freight or express bill **before** delivery is accepted or signed for by a GE Healthcare representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier immediately upon discovery or within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14-day period.

To file a report:

- Call 1-800-548-3366, and use option 8.
- Fill out a report at <http://egems.med.ge.com/edq/home.jsp>.

Contact your local service coordinator for more information on this process.

OMISSIONS AND ERRORS

Customers, contact your GE Healthcare Sales or Service representatives. GE personnel, use the Trackwise Process to report all omissions, errors, and defects in this publication.

WARNING

(EN)

This service manual is available in English only.

- If a customer's service provider requires a language other than english, it is the customer's responsibility to provide translation services.
- Do not attempt to service the equipment unless this service manual has been consulted
- Failure to heed this warning may result in injury to the service provider, operator or patient

ПРЕДУПРЕЖДЕНИЕ

(BG)

Това упътване за работа е налично само на английски език.

- Ако доставчикът на услугата на клиента изиска друг език, задължение на клиента е
- Не използвайте оборудването, преди да сте се консултирали и разбрали упътването за работа.
- Неспазването на това предупреждение може да доведе до нараняване на доставчика на услугата, оператора или пациента в резултат на токов удар, механична или друга опасност.

警告

(ZH-CN)

本维修手册仅提供英文版本。

- 如果客户的维修服务人员需要非英文版本，则客户需自行提供翻译服务。
- 未详细阅读和完全理解本维修手册之前，不得进行维修。
- 忽略本警告可能对维修服务人员、操作人员或患者造成电击、机械伤害或其他形式的伤害。

警告

(ZH-HK)

本服務手冊僅提供英文版本。

- 倘若客戶的服務供應商需要英文以外之服務手冊，客戶有責任提供翻譯服務。
- 除非已參閱本服務手冊及明白其內容，否則切勿嘗試維修設備。
- 不遵從本警告或會令服務供應商、網絡供應商或病人受到觸電、機械性或其他危險。

警告

(ZH-TW)

本維修手冊僅有英文版。

- 若客戶的維修廠商需要英文版以外的語言，應由客戶自行提供翻譯服務。
- 請勿試圖維修本設備，除非 您已查閱並瞭解本維修手冊。
- 若未留意本警告，可能導致維修廠商、操作員或病患因觸電、機械或其他危險而受傷。

UPOZORENJE

(HR)

Ovaj servisni priručnik dostupan je na engleskom jeziku.

- Ako davatelj usluge klijenta treba neki drugi jezik, klijent je dužan osigurati prijevod.
- Ne pokušavajte servisirati opremu ako niste u potpunosti pročitali i razumjeli ovaj servisni
- Zanimarite li ovo upozorenje, može doći do ozljede davatelja usluge, operatera ili

VÝSTRAHA

(CS)

Tento provozní návod existuje pouze v anglickém jazyce.

- V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka.
- Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah.
- V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obslužného personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik.

ADVARSEL

(DA)

Denne servicemanual findes kun på engelsk.

- Hvis en kundes tekniker har brug for et andet sprog end engelsk, er det kundens ansvar at sørge for oversættelse.
- Forsøg ikke at servicere udstyret uden at læse og forstå denne servicemanual.
- Manglende overholdelse af denne advarsel kan medføre skade på grund af elektrisk stød, mekanisk eller anden fare for tekniker, operatøren eller patienten.

WAARSCHUWING (NL)	Deze onderhoudshandleiding is enkel in het Engels verkrijgbaar. - Als het onderhoudspersoneel een andere taal vereist, dan is de klant verantwoordelijk voor de vertaling ervan. - Probeer de apparatuur niet te onderhouden alvorens deze onderhoudshandleiding werd geraadpleegd en begrepen is. - Indien deze waarschuwing niet wordt opgevolgd, zou het onderhoudspersoneel, de operator of een patiënt gewond kunnen raken als gevolg van een elektrische schok, mechanische of andere gevaren.
HOIATUS (ET)	See teenindusjuhend on saadaval ainult inglise keeles - Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlketeenuse osutamise eest. - Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist. - Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.
VAROITUS (FI)	Tämä huolto-ohje on saatavilla vain englanniksi. - Jos asiakkaan huoltohenkilöstö vaatii muuta kuin englanninkielistä materiaalia, tarvittavan käännöksen hankkiminen on asiakkaan vastuulla. - Älä yritä korjata laitteistoa ennen kuin olet varmasti lukenut ja ymmärtänyt tämän huolto-ohjeen. - Mikäli tätä varoitusta ei noudateta, seurauksena voi olla huoltohenkilöstön, laitteiston käyttäjän tai potilaan vahingoittuminen sähköiskun, mekaanisen vian tai muun vaaratilanteen vuoksi.
ATTENTION (FR)	Ce manuel d'installation et de maintenance est disponible uniquement en anglais. - Si le technicien d'un client a besoin de ce manuel dans une langue autre que l'anglais, il incombe au client de le faire traduire. - Ne pas tenter d'intervenir sur les équipements tant que ce manuel d'installation et de maintenance n'a pas été consulté et compris. - Le non-respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.
WARNUNG (DE)	Diese Serviceanleitung existiert nur in englischer Sprache. - Falls ein fremder Kundendienst eine andere Sprache benötigt, ist es Aufgabe des Kunden für eine entsprechende Übersetzung zu sorgen. - Versuchen Sie nicht diese Anlage zu warten, ohne diese Serviceanleitung gelesen und verstanden zu haben. - Wird diese Warnung nicht beachtet, so kann es zu Verletzungen des Kundendiensttechnikers, des Bedieners oder des Patienten durch Stromschläge, mechanische oder sonstige Gefahren kommen.
ΠΡΟΕΙΔΟΠΟΙΗΣΗ (EL)	Το παρόν εγχειρίδιο σέρβις διατίθεται μόνο στα αγγλικά. - Εάν ο τεχνικός σέρβις ενός πελάτη απαιτεί το παρόν εγχειρίδιο σε γλώσσα εκτός των αγγλικών, αποτελεί ευθύνη του πελάτη να παρέχει τις υπηρεσίες μετάφρασης. - Μην επιχειρήσετε την εκτέλεση εργασιών σέρβις στον εξοπλισμό αν δεν έχετε συμβουλευτεί και κατανοήσει το παρόν εγχειρίδιο σέρβις. - Αν δεν προσέξετε την προειδοποίηση αυτή, ενδέχεται να προκληθεί τραυματισμός στον τεχνικό σέρβις, στο χειριστή ή στον ασθενή από ηλεκτροπληξία, μηχανικούς ή άλλους κινδύνους.
FIGYELMEZTETÉS (HU)	Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el. - Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészíttetése. - Ne próbálja elkezdni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték. - Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatti sérülését eredményezheti.

AÐVÖRUN
(IS)

Þessi þjónustuhandbók er aðeins fánleg á ensku.

- Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskiptamanns að skaffa tungumálþjónustu.
- Reynið ekki að afgreiða tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilin.
- Brot á sinna þessari aðvörðun getur leitt til meiðsla á þjónustuveitanda, stjórnanda eða sjúklings frá raflosti, vélrænu eða öðrum áhættum.

AVVERTENZA
(IT)

Il presente manuale di manutenzione è disponibile soltanto in lingua inglese.

- Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione.
- Procedere alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto.
- Il mancato rispetto della presente avvertenza potrebbe causare lesioni all'addetto alla manutenzione, all'operatore o ai pazienti provocate da scosse elettriche, urti meccanici o altri rischi.

警告
(JA)

このサービスマニュアルには英語版しかありません。

- サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。
- この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

경고
(KO)

본 서비스 매뉴얼은 영어로만 이용하실 수 있습니다.

- 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우, 번역 서비스를 제공하는 것은 고객의 책임입니다.
- 본 서비스 매뉴얼을 참조하여 숙지하지 않은 이상 해당 장비를 수리하려고 시도하지 마십시오.
- 본 경고 사항에 유의하지 않으면 전기 쇼크, 기계적 위험, 또는 기타 위험으로 인해 서비스 제공자, 사용자 또는 환자에게 부상을 입힐 수 있습니다.

BRĪDINĀJUMS (LV)	Šī apkopes rokasgrāmata ir pieejama tikai angļu valodā. - Ja klienta apkopes sniedzējam nepieciešama informācija citā valodā, klienta pienākums ir nodrošināt tulkojumu. - Neveiciet aprīkojuma apkopi bez apkopes rokasgrāmatas izlasīšanas un saprašanas. - Šī brīdinājuma neievērošanas rezultātā var rasties elektriskās strāvas trieciena, mehānisku vai citu faktoru izraisītu traumu risks apkopes sniedzējam, operatoram vai pacientam.
ISPĒJIMAS (LT)	Šis ekspluatavimo vadovas yra tik anglų kalba. - Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, suteikti vertimo paslaugas privalo klientas. - Nemėginkite atlikti įrangos techninės priežiūros, jei neperskaitėte ar nesupratote šio ekspluatavimo vadovo. - Jei nepaisysite šio įspėjimo, galimi paslaugų tiekėjo, operatoriaus ar paciento sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų.
ADVARSEL (NO)	Denne servicehåndboken finnes bare på engelsk. - Hvis kundens serviceleverandør har bruk for et annet språk, er det kundens ansvar å sørge for oversettelse. - Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått. - Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.
OSTRZEŻENIE (PL)	Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim. - Jeśli serwisant klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta. - Nie próbować serwisować urządzenia bez zapoznania się z niniejszym podręcznikiem serwisowym i zrozumienia go. - Niezastosowanie się do tego ostrzeżenia może doprowadzić do obrażeń serwisanta, operatora lub pacjenta w wyniku porażenia prądem elektrycznym, zagrożenia mechanicznego bądź innego.
AVISO (PT-BR)	Este manual de assistência técnica encontra-se disponível unicamente em inglês. - Se outro serviço de assistência técnica solicitar a tradução deste manual, caberá ao cliente fornecer os serviços de tradução. - Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica. - A não observância deste aviso pode ocasionar ferimentos no técnico, operador ou paciente decorrentes de choques elétricos, mecânicos ou outros.
ATENÇÃO (PT-PT)	Este manual de assistência técnica só se encontra disponível em inglês. - Se qualquer outro serviço de assistência técnica solicitar este manual noutra idioma, é da responsabilidade do cliente fornecer os serviços de tradução. - Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica. - O não cumprimento deste aviso pode colocar em perigo a segurança do técnico, do operador ou do paciente devido a choques eléctricos, mecânicos ou outros.
ATENȚIE (RO)	Acest manual de service este disponibil doar în limba engleză. - Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere. - Nu încercați să reparați echipamentul decât ulterior consultării și înțelegerii acestui manual de service. - Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.
ОСТОРОЖНО! (RU)	Данное руководство по техническому обслуживанию представлено только на английском языке. - Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод. - Перед техническим обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения. - Несоблюдение требований данного предупреждения может привести к тому, что специалист по техобслуживанию, оператор или пациент получит удар электрическим током, механическую травму или другое повреждение.

- UPOZORENJE**
(SR)
- Ovo servisno uputstvo je dostupno samo na engleskom jeziku.
- Ako klijentov serviser zahteva neki drugi jezik, klijent je dužan da obezbedi prevodilačke usluge.
 - Ne pokušavajte da opravite uređaj ako niste pročitali i razumeli ovo servisno uputstvo.
 - Zanemarivanje ovog upozorenja može dovesti do povređivanja serviser, rukovaoca ili pacijenta usled strujnog udara ili mehaničkih i drugih opasnosti.
- UPOZORNENIE**
(SK)
- Tento návod na obsluhu je k dispozícii len v angličtine.
- Ak zákazník poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka.
 - Nepokúšajte sa o obsluhu zariadenia, kým si neprečítate návod na obsluhu a neporozumiete mu.
 - Zanedbanie tohto upozornenia môže spôsobiť zranenie poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, mechanické alebo iné ohrozenie.
- ATENCION**
(ES)
- Este manual de servicio sólo existe en inglés.
- Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual.
 - No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio.
 - La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.
- VARNING**
(SV)
- Den här servicehandboken finns bara tillgänglig på engelska. .
- Om en kunds servicetekniker har behov av ett annat språk än engelska, ansvarar kunden för att tillhandahålla översättningstjänster.
 - Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken.
 - Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.
- DİKKAT**
(TR)
- Bu servis kılavuzunun sadece ingilizcesi mevcuttur.
- Eğer müşteri teknisyeni bu kılavuzu ingilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer.
 - Servis kılavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz.
 - Bu uyarıya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.

TABLE OF CONTENTS

	DAMAGE IN TRANSPORTATION	3
	OMISSIONS AND ERRORS	3
1	INTRODUCTION	10
1-1	Functool 9 Advanced Application “Clinical” Summary	10
2	PRELIMINARY REQUIREMENTS	10
2-1	License Key Number	10
2-2	Save Configuration	11
2-2.1	Reading Config Media → PC	11
2-2.2	Reading Config Media → AW	11
3	INSTALLATION PROCEDURE	11
4	AW AND FUNCTOOL 9 ERRORS	12
5	ADVANCED SOFTWARE APPLICATION OPERATION AND DEMO EXAM INFORMATION	12
6	INSTALLATION FINISH TASKS	12
7	SPECIAL CONSIDERATIONS	12
8	FRU LIST	13

1 INTRODUCTION

The following manual gives a brief overview of the installation process for the Functool 9 advanced application software option. The Functool 9 application is designed to run on the AW4.1 and forward AW platforms only. This guide makes references to all the appropriate standard Advantage Workstation service manuals for more detailed information about installing an advanced application software option on the AW. The current revision of the appropriate AW Service Manual should be used when installing this software. Unique installation processes for Functool 9 and NOT found in the AW service manual can be found in this document under the section entitled SPECIAL CONSIDERATIONS.

1-1 Functool 9 Advanced Application “Clinical” Summary

Functool 9 is an application that is used to analyze Magnetic Resonance Imaging (MRI) data sets that contain multiple images for each scan location, representing changes in intensity over time (“time series” or “dynamic” data sets), b-value (diffusion imaging), gradient orientation (diffusion tensor imaging) or frequency (spectroscopy).

Functool 9 provides the user with two main tools for analysis:

- Graphs – these show the changes in pixel value over time, b-value or frequency at a given point on the images, either at the cursor position or within a region of interest (ROI) defined by the user.
- Functional maps – the pixel values from the data set are used to compute a characteristic parameter for every pixel location by means of a function. This can be a basic algorithm or a function derived from such an algorithm. A functional map is then constructed by displaying the value of this parameter for every pixel location.

Functool 9 is supplied in the form of protocols. Each consists of one or more protocol panels that contain the necessary information and the controls required to set up an analysis.

2 PRELIMINARY REQUIREMENTS

2-1 License Key Number

The Functool 9 option requires a license key # based upon the workstation’s Host ID. This key number may be provided by:

- Paperwork shipped with the option that will include one or more 8- to 10-digit license key number(s) – created based on the workstation’s/application’s FDO and the workstation’s Host ID/License ID.
- The software package may include a portable media (i.e. floppy disk) with the license key information on it.
- The license key information will also be available online at the eLicense website link by searching under the order number or the workstation’s Host ID/License ID:
<http://elicense.gehealthcare.com>.

All of these elements may exist (hard copy license #(s), media, and eLicense data), and at least one of them is necessary to load the software.

2-2 Save Configuration

Create an AW 'CONFIG' media before and after the application software installation for use as needed throughout this and other AW service activities.

2-2.1 Reading Config Media → PC

Configuration information can be read on a PC by viewing the config.txt file on the media using text readers like WordPad, Internet Explorer or MS Word (MS NotePad may not work).

2-2.2 Reading Config Media → AW

To read a Save Configuration using a floppy disk insert the floppy and type the following in a c shell:

- **su**
- Password '**operator**'
- **mount /mnt/floppy**
- **cd /mnt/floppy**
- **more config.txt**
- **cd /**
- **umount /mnt/floppy**

To read a Save Configuration using a CDROM disk insert the CD and type the following in a c shell:

- **su**
- Password '**operator**'
- **mount /mnt/cdrom**
- **cd /mnt/cdrom**
- **more ("backup directory name")**
- **cd /**
- **umount /mnt/cdrom**

3 INSTALLATION PROCEDURE

The 'Standard' AW Software Options Installation – Easy Install process applies. For ongoing process revision viability, the details will not be printed here, but they can be found in the AW Service Manual.

The current revisions for this manual can be viewed online at:

<http://aw-ib.euro.med.ge.com>.

1. Open up the appropriate Advantage Workstation (AW) Service Manual.
2. Proceed to Job Card entitled – Software Options Installation.
3. Follow the procedure using the "Easy Install" method.
4. When complete, perform a save configuration.

4 AW AND FUNCTOOL 9 ERRORS

Functool 9 and AW System errors can be examined and investigated in either the usual AW error logs – such as /var/adm/messages, and/or ~sdc/logfiles/easyInstallLog. Errors specific to Functool 9 can be found in /export/home/sdc/logfiles/functool.log.

5 ADVANCED SOFTWARE APPLICATION OPERATION AND DEMO EXAM INFORMATION

To test the functionality of the software, first you must load the DEMO exam CD shipped with the product. The 'Standard' AW Software Demo Exam Installation process applies. For ongoing process revision viability, the details will not be printed here, but they can be found in the AW Service Manual.

The current revisions for this manual can be viewed online at:
<http://aw-ib.euro.med.ge.com>.

1. Open up the appropriate Advantage Workstation (AW) Service Manual.
2. Proceed to Job Card entitled – Advantage Workstation Demo Exam Reinstallation.
3. Follow the procedure to load the Demo Exams from the CD provided.

To test this application, after the application is installed, logout and log back into the AW (as suggested by the messages after the installation). Then select any data set and click on Functool in the "Applications" menu. The Functool window will open (to the left side of the screen). This ensures that Functool is installed correctly. Click on the Quit button to exit Functool.

6 INSTALLATION FINISH TASKS

GIB Update - The Functool 9 application must be entered into GIB (Global Installed Base) database after the initial successful installation in order to fulfill product-tracking requirements.

FAILURE TO ENTER THE SYSTEM AND ALL SOFTWARE APPLICATIONS INTO GIB will result in GE Healthcare being in violation of FDA tracking compliance, and this system will not be in the affectivity list for future upgrades.

The OLC AW Support line can direct you as how to get GIB updated correctly if you don't know. Here is the global GIB link:

GIB: <http://gib.gehealthcare.com>

7 SPECIAL CONSIDERATIONS

In addition to Functool, Functool CSI, Functool DTI, Functool FiberTrak, Functool BrainStat AIF, Functool Fusion, Functool eDWI, Functool MR-Touch and Functool 3DASL will show up at the package installation time as separate license key entries.

In addition to the required, Functool license key, the following optional license keys are available:

- Functool CSI
- Functool DTI
- Functool FiberTrak
- Functool BrainStat AIF
- Functool Fusion
- Functool EDWI
- Functool MR-Touch
- Functool 3DASL

If any of these options was purchased, each will have a license key. The license keys must be available during installation. The purchased options must be selected and the license key numbers entered during installation. At the installation key prompt, If the option was not purchased, leave it unselected.

8 FRU LIST

The replacement parts for this option are as follows:

Part# 5198959-3 – Functool 9 Software and Documentation CD-ROM.